

4. ard_K210 mechanical code identification

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1.K210 and Arduino communication

1.1 Experimental premises

This tutorial uses arduino, and K210 requires running the program in **K210-AI (stm32_pico_arduino)** to start the experiment

arduino *1

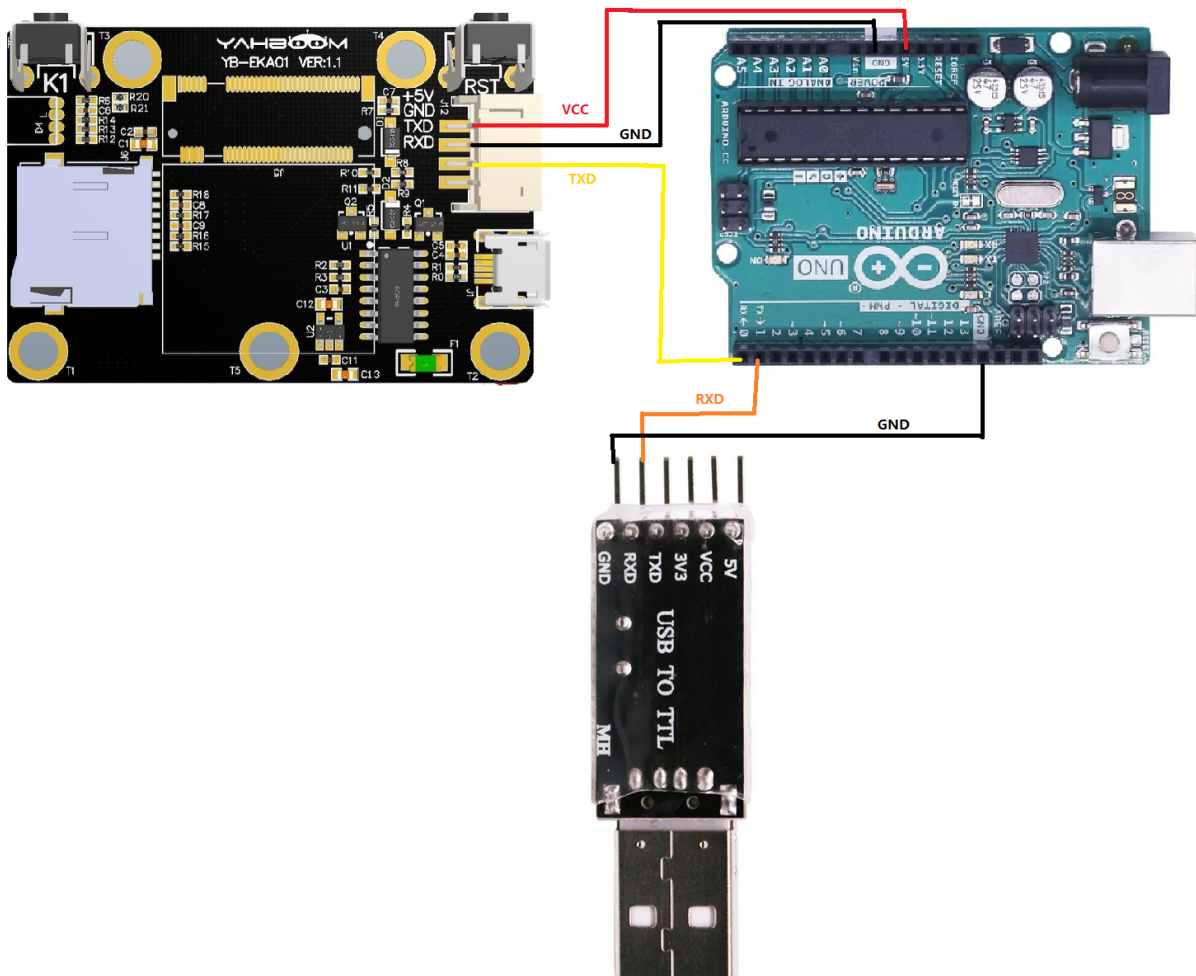
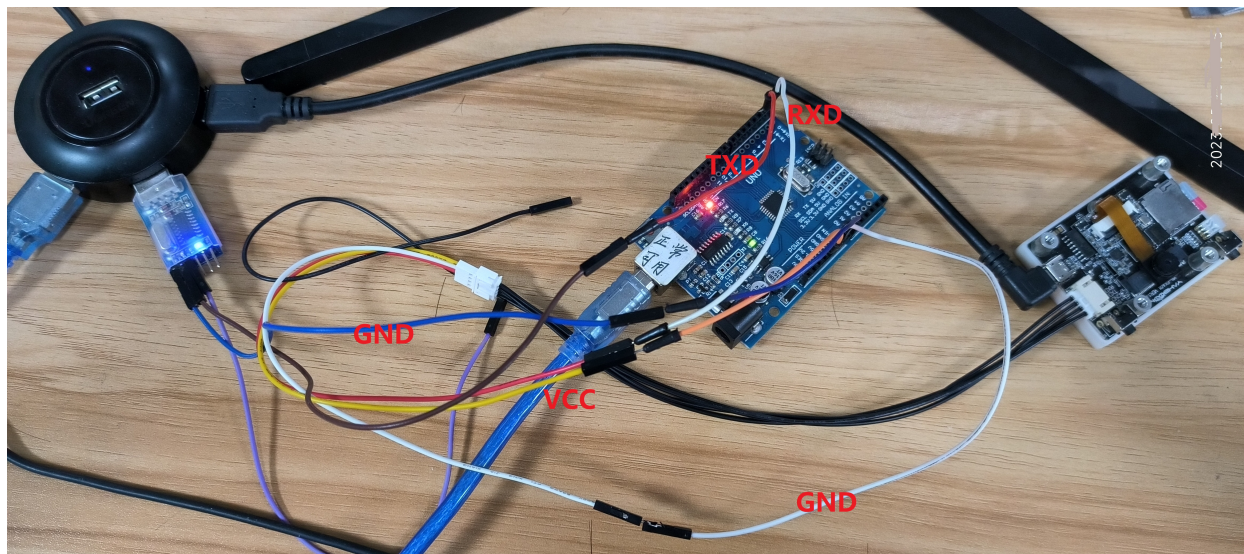
K210 perspective module * 1 (requires SD card (with AI model inside) and camera)

USB to TTL module * 1

1.2 Experimental wiring

arduino	usb to ttl
TXD	RXD
GND	GND

arduino	K210 perspective module
RXD	TXD
GND	GND
VCC	5V
Wiring as shown in the diagram:	



1.3 Main code explanation

```
void loop()
{
    while (K210Serial.available())
    {
```

```

recv_k210msg(k210Serial.read());

if (k210_msg.class_n != 0)
{
    if(k210_msg.class_n == 4)
    {

        sprintf(buff_com, "x=%d,y=%d,w=%d,h=%d\r\n", k210_msg.x, k210_msg.y, k210_msg.w, k210_msg.h);

        k210Serial.print(buff_com);

        sprintf(buff_com, "id = %c%c, str = %s\r\n",
(k210_msg.id>>8), k210_msg.id, k210_msg.msg_msg);
        k210Serial.print(buff_com);

        k210_msg.class_n = 0;
    }

}

}
}

```

After the above program, if you are running this routine, k210_ The members of the msg structure have corresponding values and are processed through serial port printing
K210_Msg: is a structure that receives information, and its main members are

- X: is the horizontal coordinate of the top left corner of the recognized box (range: 0-240)
- Y: is the vertical coordinate of the upper left corner of the identified box (range: 0-320)
- W: is the width of the recognized box (range: 0-240)
- H: The length of the recognized box (range: 0-320)
- ID: is the recognized label
- Class_n: Routine number
- Msg_Msg [20]: Valid data

After receiving and processing data, k210_ Each member of the msg will store valid information.
If you want to develop it again, call K210 directly_ Members of msg are sufficient

1.4 experimental phenomena

1. After connecting the cable, the K210 perspective module runs offline. Please check 【6.2 K210 as coprocessor】 -- 【ReadMe】

2. Set the serial port assistant to the interface shown in the figure

The image shows a serial port configuration window. The 'COM Configs' section has the following settings: Channel: COM1, Baudrate: 115200, Paritybits: NONE, Databits: 8, Stopbits: 1, Flowctrl: NONE. There is a 'Close' button with a red cross icon. The 'Recv Options' section has: ASCII selected, Log display mode checked, Auto linefeed checked, Hide received data unchecked, Recv save to file... unchecked, and links for AutoScroll and Clear. The 'Send Options' section has: ASCII selected, Enable escape chars checked, AT CMD auto CRLF checked, Auto append bytes unchecked, Send from file ... unchecked, Period: 1000 ms, and links for Shortcut and History.

3. Then run the mechanical code recognition routine, and the serial assistant will print out the important information transmitted from k210 to stm32, as shown in the following figure

The image shows a serial terminal window with a black background and green text. The text displays a series of data points transmitted from k210 to stm32. Each line follows the format: id = [value], str = TAG16H5, x=[value], y=[value], w=[value], h=[value]. The data points are as follows:

```
id = 01, str = TAG16H5
x=112, y=17, w=23, h=23
id = 01, str = TAG16H5
x=112, y=18, w=23, h=22
id = 01, str = TAG16H5
x=111, y=19, w=23, h=22
id = 01, str = TAG16H5
x=109, y=19, w=24, h=22
id = 01, str = TAG16H5
x=109, y=18, w=23, h=23
id = 01, str = TAG16H5
x=106, y=19, w=23, h=23
id = 01, str = TAG16H5
x=107, y=20, w=23, h=22
id = 01, str = TAG16H5
x=106, y=19, w=23, h=22
id = 01, str = TAG16H5
x=104, y=13, w=23, h=22
id = 01, str = TAG16H5
x=101, y=4, w=24, h=23
id = 01, str = TAG16H5
x=102, y=7, w=24, h=23
id = 01, str = TAG16H5
x=103, y=9, w=23, h=23
id = 01, str = TAG16H5
```

Mechanical code recognition only transmits the six Member variable of k210_msg, namely, x, y, w, h, msg and id.

